

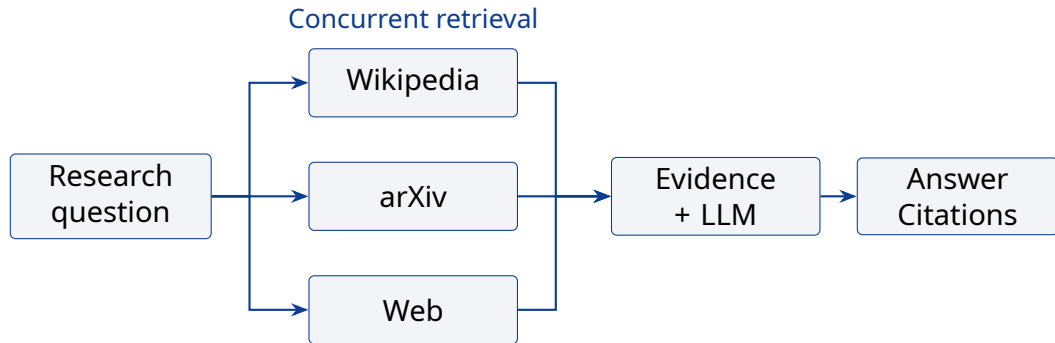
AI Academy, National AI Center

# Research Assistant

**VerteX**

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# One question, multiple sources

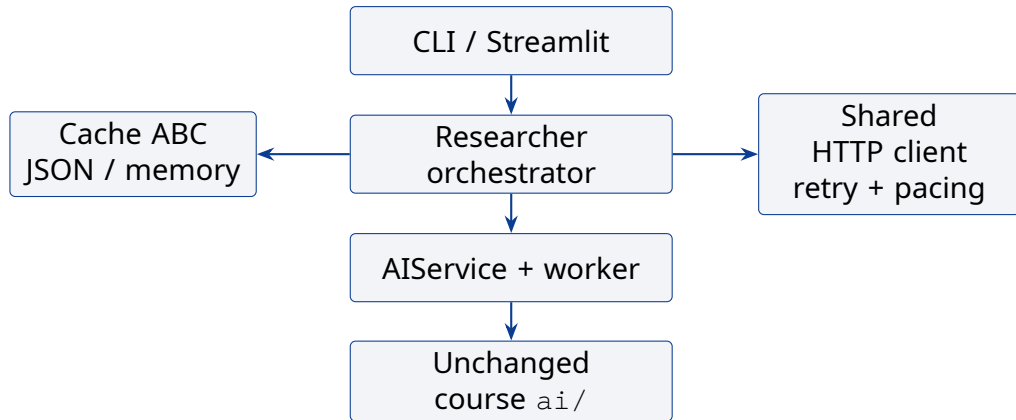


CLI or Streamlit

Source status

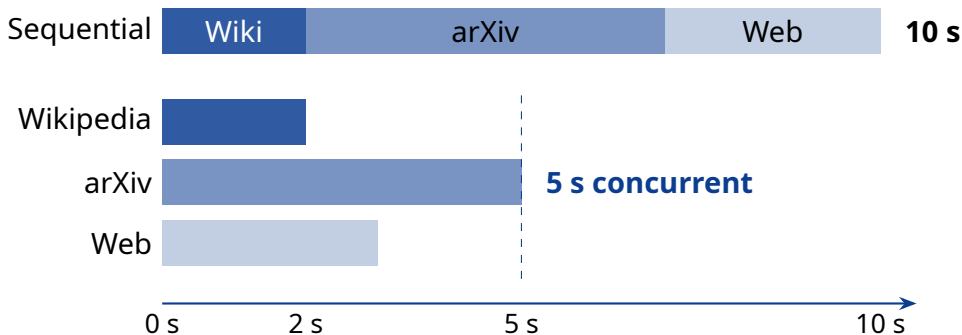
Text or JSON

# Architecture



Pydantic models validate data across boundaries.

# Why async retrieval?



Illustrative timings. Semaphore limits active work. LLM synthesis follows retrieval.

# Measured retrieval speedup

Sequential



Concurrent



**3.01×**

retrieval speedup

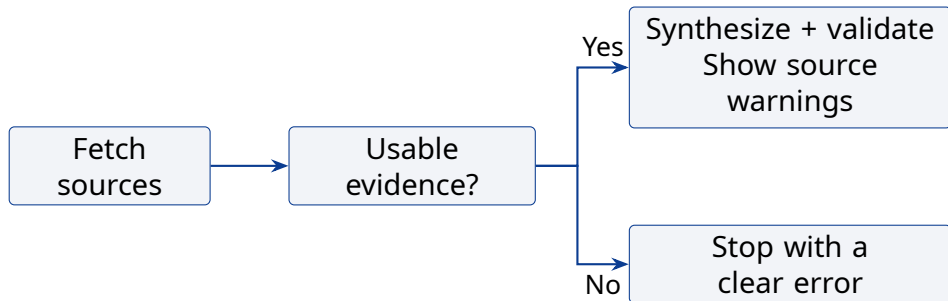
5 questions

3 repetitions

0 failed / empty

Simulated HTTP: 150 ms per source. Median times, cache bypassed.  
Retrieval only. These results do not measure live LLM latency.

# Failure handling



**Slow source**

Deadline bounds the wait

**Uncited answer**

Regenerate within the retry budget

# Demonstration modes

## Offline sample

5 prepared topics

No network or API key

Explicitly simulated evidence

## Live research

External source requests

Configured LLM key

Latency and quota vary

```
python -m researcher demo --offline --no-cache
```

Live demo: inspect references and source status in Streamlit.

## Verification evidence

**121**

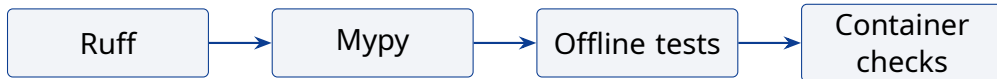
tests passed

**93.85%**

application coverage

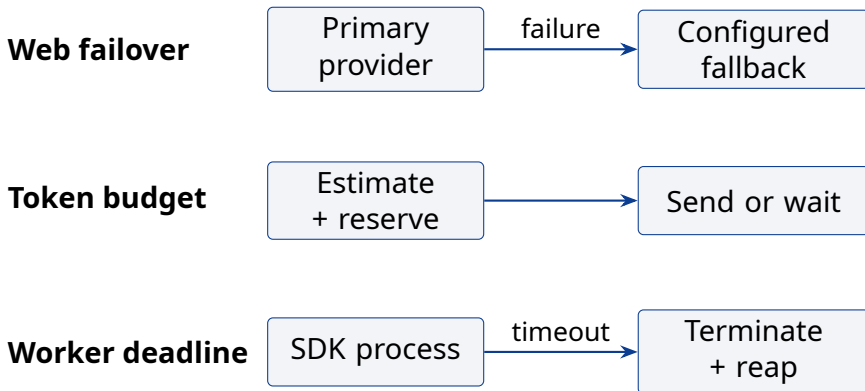
**16**

course smoke tests



Recorded full run: 17 September 2026. Smoke tests are included in 121. Tests mock external services. Coverage measures execution, not correctness.

# Reliability extensions



Failover needs configured providers. Token estimates apply to one service instance.

# Live demonstration

Research Assistant

Scan to open the deployed app.

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Questions and discussion

